Systems and Methods for Aircraft Supplemental Oxygen Control

Background:

Supplemental oxygen in aviation in the U.S. is regulated by the Code of Federal Regulations (14 CFR 91.211) to prevent hypoxia (oxygen deprivation) at high altitudes. Other countries and jurisdictions have similar regulations to assist with the safe operation of aircraft at high altitudes. The configurations described herein are intended to assist in improving the delivery and control of critical oxygen resource to pilots and other aircraft occupants in various types of aircraft which may encounter high altitude scenarios.

Referring to Figure 1A, one of the most popular general aviation single engine aircraft of all time, known as the Cessna 172 (2), is shown. The Cessna 172 is known for safety and stability, and has been utilized for generations as a training aircraft since its launch in the mid 1950s. Figure 1B illustrates aspects of a dashboard control interface (4) of a typical Cessna 172 (2) or other conventional single-engine general aviation aircraft, with two operational/steering yokes (8, 10), and various gauges (6), some of which are configured to illustrate aspects critical to safe operation, such as altitude, speed, horizon orientation, and other aspects, as well as communications information and configuration interfaces (such as via a VHF communications system or radio). Referring to Figure 2A, a more modern single engine aircraft known as the Cirrus

SR22 (12) has become one of the most popular general aviation aircraft in history since its launch in 2001, and has become known for its safety features and advanced controls. Referring to Figure 2B, an interior view of an SR22 (12) shows a dash assembly (14) featuring left (20) and right (22) yokes or control interfaces, a plurality of flat panel displays (16), as well as many control and/or informational buttons (18), including controls for communications such as via a VHF communications system or radio configuration. Referring to Figures 2C and 2D, also positioned within the interior of the SR22 is a mechanical interface (24) for causing a safety parachute (26) to be released from the fuselage of the SR22, at the option of one of the occupants of the aircraft, so that it may form an expanded condition as shown in Figure 2D and facilitate safe landing of the aircraft (12) in the event of an emergency such as fuel problem, engine problem, pilot incapacity, or other scenario wherein safe landing would likely otherwise be risky if not impossible. Such feature has purportedly been involved in the saving of hundreds of lives since the product was introduced, and indeed has been a welcomed safety feature for such aircraft. Referring to Figure 2E, more recently, a so-called "Safe Return™" feature has been introduced to certain Cirrus™ aircraft which allows occupants of aircraft with the feature to press a button (28) on the interior (30) of the aircraft to engage such feature, which automatically coordinates with air traffic control ground-based resources, as well as automated computerized resources, to safety land the aircraft at a nearby airport in the case of an emergency (such as when a pilot becomes incapacitated and other occupants do not know how to fly and/or land the aircraft). Such feature also has been welcomed by many as a valued safety feature which will no doubt save many lives.

Referring to Figures 3A, 3B, and 3C, various configurations for delivering supplemental oxygen to a pilot (32) or other aircraft occupant (33) are illustrated for scenarios wherein, typically at high altitudes above approximately 10,000 feet, supplemental oxygen is recommended to ensure safety and comfort of the pilot (32) and other occupants (33) given the proportion of oxygen in the air available from the environment at such altitudes. Typically when the pilot is beginning to gain altitude via operation or the plane to an altitude wherein supplemental oxygen is recommended or required, and pilot (32) and occupants (33) will wear an oxygen delivery mask, tube, or other distribution apparatus, and will open a valve (such as mechanical/rotary valve on an oxygen container or tank) to provide flow of oxygen to the nose and/or mouth of the use for intake respiration. Figure 3A illustrates a tubular (34) style configuration with outlets at the nostrils of the user (32); Figure 3B illustrates a strapped on partial mask assembly (36) with outputs adjacent the nostrils of the users (32, 33); Figure 3C illustrates strapped on more complete mask assemblies (38) configured to provide oxygen flow to inhalation of the users (32, 33) via the mouth or the nostrils, and to provide a greater sealing engagement between the supplemental oxygen supply and the user in general. Referring to Figures 4A, 4B, and 4C, one or more oxygen tanks (42) and associated oxygen distribution tubes or conduits (40) may be stored and/or positioned for such in-flight supplemental oxygen purposes as a tank/valve assembly mounted to a portion of the aircraft interior such as the front seat (as shown in Figure 4A), rear seat (48) as shown in Figure 4C, or fuselage (46) as shown in Figure 4B.

Unfortunately, from time-to-time, there are in-flight problems with the delivery of necessary oxygen to a pilot or other aircraft occupants. When a pilot or other occupant fails to have enough oxygen (a condition known as "hypoxia"), they typically start to feel dizzy or lightheaded and may lose consciousness without much notice (indeed, in some scenarios, they may have very little warning). When a person sitting on a couch in a home enters a state of hypoxia for whatever reason, there certainly is a risk of bad outcomes for that person, but given their position seated stably on a couch in a home, such condition may not have a lot of risk of injury to others, and hopefully such person can be exited out of such state of hypoxia in a hurry so that they can remain stable from a consciousness perspective. When a pilot (32) or other occupant (33) of an aircraft at altitude and relatively high velocity enters a hypoxia scenario, there arguably is significant risk for the pilot (32), occupants (33), and others who may be in the path of the aircraft. Also unfortunately, as noted above, hypoxia seems to "sneak up" on people such that there is not a lot of notice to them before they are incapacitated or unconscious. In such cases, there arguably is not enough time, for example, to pull to actuate an aircraft rescue parachute (as described above in reference to Figures 2C and 2D), or to press a button to execute a safe automatic landing (as described above in reference to Figure 2E). Indeed, with such scenarios, it may be the case that some or all of the persons at altitude without sufficient oxygen not only become unconscious or incapacitated, but that they become deceased from lack of sufficient oxygen. Referring to Figure 5A, a photograph of an aircraft flying at high altitude with frosted over windshields is shown; tragically such aircraft was found to be flying on autopilot at high altitude with all occupants suffering from hypoxia, probably

deceased before the aircraft ultimately ran out of fuel and crashed, as shown in Figure 5B. The intentions of the configurations described herein are to try to prevent such accidents from ever happening again by providing sophisticated controls and warnings so that hypoxia may be prevented and quickly addressed in critical situations.

DETAILED DESCRIPTION:

Referring to Figures 6A-6C, various pulse oximeter configurations are illustrated which may be purchased by consumers for various applications such as mountain climbing. Figure 6A illustrates a configuration made to receive a finger (52) of the hand (50) of a user between two lightly-spring-loaded opposing portions (66, 67) while a controller emits a small amount of light radiation through a portion of the finger (52) to calculate the partial pressure of oxygen in the user's bloodstream. Figure 6B illustrates a somewhat similar pulse oximeter configuration configured to fit around a thumb (54) of a user's hand (50) to provide the partial pressure of oxygen in the user's bloodstream with such a compact geometry. Figure 6C illustrates a configuration which may be most conducive to normal operation of the hand (50) of the user, wherein a small and light-weight reader cuff assembly (64) may be fastened around a finger or thumb of the user's hand (50), the cuff assembly (64) operatively coupled (such as via a wire or other signal lead 62) to wrist-based (akin to a watch) assembly having a controller or process and display, which may be coordinated to provide visible readings regarding the partial pressure of oxygen in the user's bloodstream. Each of these systems may be configured to output data, such as real-time or near-real-time readings of the partial pressure of oxygen in the user's bloodstream via wired or wireless connectivity (such as BluetoothTM, IEEE 802.11, Near-Field, or other wireless connectivity paradigm) to other connected systems (such as smartphones, tablet computers, laptops, or other systems).

Referring to Figure 7A, in one embodiment, a closed-loop oxygen management system is illustrated wherein a person (68), such as an aircraft pilot or other occupant, is wearing an oxygen supply mask (34; such as that described above in reference to Figure 3C) as well as a pulse oximeter (such as that described in reference to Figure 6C, elements 60, 64, 62; element 76 illustrates an RF transceiver, such as one which may be utilized to wirelessly communicate or connect with other resources, such as via Bluetooth™, IEEE 802.11, Near-Field, or other configuration; such a configuration facilitates relatively free operation of the hand for a pilot or other person; other configurations such as those described in Figures 6A and 6B may also be utilized) operatively coupled to the hand (50) of the person (68). In this relatively minimal system configuration, an oxygen tank or reservoir (42) featuring an intercoupled flow controller assembly (78) and associated RF transceiver (74; such as one which may be utilized to wirelessly communicate or connect with other resources, such as via Bluetooth™, IEEE 802.11, Near-Field, or other configuration) may be configured to have an oxygen output conduit (40) coupled between the flow controller assembly (78) and oxygen mask (38); further the flow controller assembly (78) may be operatively coupled to a local computing resource (70; comprising a device such as a microprocessor or microcontroller) using a similar wireless communications transceiver (72; such as via Bluetooth™, IEEE 802.11, Near-Field, or other configuration), and the pulse oximeter (60) may be operatively coupled, such as via wireless communication as noted, to the local computing resource (70). With the pulse oximeter (60), local computing resource (70), and flow controller (78) operatively coupled (again, such as via wireless communication 76, 72, 74), a code base may be provided, such as on a memory device accessible to

the local computing resource (70; such as via stable nonvolatile flash memory or other stable electronic storage) and operated by such local computing resource to create an important fundamental closed loop control paradigm whereby the person (68) is adequately supplied (with reference to confirming pulse oximeter 60 data from the person) with supplemental oxygen via the flow controller (78), as coordinated by the local computing resource (70). If the person's (68) partial pressure of oxygen level, as calculated using the pulse oximeter (60) goes below a threshold amount, the local computing resource (70) may be configured to cause the flow controller (78) increase output flow of oxygen through the conduit (40) to the person (68) via the mask (38). For example, in various embodiments, should the person's partial pressure of oxygen reading go below 90% for a specified period of time, or should the person's partial pressure of oxygen reading decrease by more than 5% below baseline for a specified period of time after a steady baseline has been otherwise determined, flow of oxygen may be automatically increased; should such increase not bring the person out of these conditions (less than about 90%; or below baseline by about 5%), the scenario may be configured to be automatically escalated (such as via an even larger increase in oxygen flow, notifications, alarms, and contacts to person within and/or outside of the aircraft, and/or other responses). In other words, in various scenarios, should the person (68) not be able to be retained within a given acceptable zone of oxygenation as per the pulse oximeter (60), the local computing resource may be configured to create an alarm or notification to be received by the particular person or others. Further, the system, such as via the local computing resource (70) and intercoupled flow controller (78), may be configured to observe oxygen flow rates and reservoir (42) capacity to assist in

providing warnings and/or notifications to the person (68) or others regarding estimated time remaining on the requisite oxygen flow, so that the pilot or other person may plan ahead to avoid potential problems.

In another more-minimal variation, the pulse oximeter may also be operatively coupled (again, such as via wireless 76, 78) directly to the flow controller (78), and a microcontroller or microprocessor resident within the pulse oximeter assembly (60) may be configured to provide the closed loop control for such system without additional computing (such as via a local computing resource 70).

Referring to Figure 7B, a variation similar to that of Figure 7A is illustrated with further intercoupled components, such as a local smartphone (82) or other local mobile computing device which may also have wireless communications capability via a transceiver (80; such for via Bluetooth™, IEEE 802.11, Near-Field, or other configuration), and such as a ground-based computing assembly (86), which may be operatively coupled to air traffic control and other key resources, such as via wireless communications and transceiver (88; such as via various RF communications technologies utilized by aircraft and ground-based resources; the transceiver 72 of the local computing resource 70 may be further configured to utilize such communications to be in communication and/or control with the ground-based computing assembly 86; alternatively the local computing resource 70 may be operatively coupled to the communications resources of the particular aircraft (such as those commonly controllable via the dash controls of the particular aircraft, such as one or more VHR communications systems or radios), which themselves generally will be coupleable to the ground-based communications and/or computing

resources (86, 88). In other words, certain aircraft-based devices or systems (such as a pulse oximeter 60, flow control valve 78, smartphone 82, and/or local computing resource 70) may be configured to communicate with each other using relatively low-power communications paradigms (such as Bluetooth™, IEEE 802.11, Near-Field, or other configuration), some or all of these devices (such as the local computing resource 70) also may be configured to utilize relatively higher-power communications configurations, such as integration with the onboard VHF communications system of the aircraft, to communicate with ground-based resources (86). Note also that certain specific devices, such as a smartphone (82), may be able to communicate with certain ground-based resources (such as 86) via utilization of cellphone / mobile connectivity networks and/or GPS or satellite connectivity.

The local smartphone or other local mobile computing device (82) may be configured to simply observe the control paradigm (for example, to receive notifications or alarms, display data, communicate with other devices such as via BlueToothTM, IEEE 802.11, Near-Field, cellular connectivity, and/or satellite connectivity), or may be configured to play a role in the control loop, such as by operating as a microcontroller or microprocessor node in the control loop. Indeed, referring to Figure 7C, an embodiment is illustrated wherein the local smartphone or other local mobile computing device (82) may be utilized to functionally replace the local computing resource (70) capability shown in the embodiment of Figure 7B, for example, and to generally serve as a communications hub between the pulse oximeter (60) and oxygen flow valve (78) - as well as ground-based computing resources (86; such as via cellular/mobile connectivity, integration with aircraft-based

VHF communications which are in communication with ground-based resources, and/or GPS or satellite communication which may be utilized to connect various disparately-located resources).

Referring to Figure 7D, although somewhat physically inconvenient relative to wireless connectivity, various components of any of such system embodiments (with exception of remotely-positioned ground-based resources 86) may be physically and operatively coupled using, for example, wired leads (90, 92, 94), and in some variations, for enhanced security, both wired and wireless communications may be utilized for redundancy.

Referring to Figure 8, the interconnectivity (again, such as via wireless and/or wired communication configuration) of various components is diagrammed for various embodiments. For example, referring to Figure 8, an oxygen saturation device such as a pulse oximeter (60; other associated components may comprise 62, 64, for example) may be operatively coupled (108) to the oxygen flow controller (42), to (104) a mobile computing resource (82), and to (102) a local computing resource (70). The oxygen flow controller (42) may be operatively coupled to (110) the mobile computing resource (82) and to (106) the local computing resource (70). The local computing resource (70) may be operatively coupled to (112) the mobile computing resource (82). An aircraft control and communications computing resource (96; such as a dash-mounted communications and mapping assembly resident in the particular aircraft, as noted above) may be operatively coupled to (114) the local computing resource (70) as well as to (118) the mobile computing resource (82) and to (120) remote ground-based communications and computing resources (86; may include air traffic control and other resources). The remote ground-based communications and computing resources (86) may be operatively coupled to (116) the local computing resource

(70), to (122) the mobile computing resource (82), or other resources in the control loop (such as directly or indirectly to the oxygen flow controller 42, such as via the local computing resource 70).

Referring to Figure 9, in one embodiment, on-board operatively-coupled computing devices and modules (such as local computing resource, mobile computing resource, oxygen saturation device, oxygen controller, aircraft control system) may be powered up and working with oxygen saturation device coupled to person (such as aircraft pilot) (130). Remote computing and communications resource (may include aspects of Air Traffic Control) may be operatively coupled and working with one or more on-board computing devices (132). Local and/or mobile on-board computing resource may be configured to monitor oxygen saturation of coupled person as well as appropriate operation of oxygen controller (and intercoupled oxygen source) given designated closed-loop control paradigm configured to avoid risk, surprise, or accident via a control loop (may involve notifications, alarms, warnings, automatic changes in oxygen flow, contact and/or communications dispatches with Air Traffic Control, and other automated actions; may be configured to sequence actions, such as via increased oxygen flow, then alarms and communications, etc - such as in the description above in reference to Figure 7A with regard to the example embodiment wherein actions may be based upon an oxygen saturation level which moves below 90%, or below an established baseline by 5%, for a specified period of time) (134). Continued safe operation of the subject aircraft, with continued mitigation of hypoxia risk due to persistent monitoring and available automated intervention, may be facilitated (136).

Referring to Figure 10, in another embodiment, a pilot may board an aircraft and begin a checklist prior to engine start; the plan may be to fly at altitude which will require supplemental oxygen (140). A local aircraft control system may be configured to not only control and/or monitor engine, communications, and other flight and/or navigation features, but also may be operatively coupled to devices such as an oxygen saturation device and an oxygen controller to assist with providing supplemental oxygen in appropriate conditions (142). A pilot checklist exercise may comprise checking not only flight control, communications, and other conventional features, but also supplemental oxygen configuration and configuration and operation of other associated and/or intercoupled devices and components (144). Upon successful checking, testing, and/or confirmation, engine start may progress (146). The pilot may proceed with takeoff and flight, and may navigate to altitude where supplemental oxygen should be initiated, at which point the supplemental oxygen control system may be configured to notify the pilot and/or other resources to position an oxygen mask or delivery system, and begin controlled oxygen delivery under closed loop control (such as via utilization of the oxygen saturation device and oxygen controller) (148). In the event of an unexpected situation pertaining to supplemental oxygen, the aircraft control system, as well as operatively coupled ground-based resources, may be utilized to assist (150); for example, the system may be configured to sequence actions, such as via increased oxygen flow, then alarms and communications, etc - such as in the description above in reference to Figure 7A with regard to the example embodiment wherein actions may be based upon an oxygen saturation level which moves below 90%, or below an established baseline by 5%, for a specified period of time. For example, in various embodiments, such resources may be

configured to create unavoidable alarms, warnings, and/or notifications for the pilot and other passengers, to force or automate reduction in altitude, to contact external resources such as air traffic control, to allow for outside intervention into the control of the aircraft (such as in a scenario wherein air traffic control or other resources may be utilized to cause a controlled decrease in elevation, change in direction, remote flight operation, to test certain connected components of the oxygen control system or other components or controls of the aircraft, or other procedures to assist in safe operation of the aircraft and safe landing for the occupants).

Various exemplary embodiments of the invention are described herein. Reference is made to these examples in a non-limiting sense. They are provided to illustrate more broadly applicable aspects of the invention. Various changes may be made to the invention described and equivalents may be substituted without departing from the true spirit and scope of the invention. In addition, many modifications may be made to adapt a particular situation, material, composition of matter, process, process act(s) or step(s) to the objective(s), spirit or scope of the present invention. Further, as will be appreciated by those with skill in the art that each of the individual variations described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present inventions. All such modifications are intended to be within the scope of claims associated with this disclosure.

The invention includes methods that may be performed using the subject devices. The methods may comprise the act of providing such a suitable device. Such provision may be

performed by the end user. In other words, the "providing" act merely requires the end user obtain, access, approach, position, set-up, activate, power-up or otherwise act to provide the requisite device in the subject method. Methods recited herein may be carried out in any order of the recited events which is logically possible, as well as in the recited order of events.

Exemplary aspects of the invention, together with details regarding material selection and manufacture have been set forth above. As for other details of the present invention, these may be appreciated in connection with the above-referenced patents and publications as well as generally known or appreciated by those with skill in the art. The same may hold true with respect to method-based aspects of the invention in terms of additional acts as commonly or logically employed.

In addition, though the invention has been described in reference to several examples optionally incorporating various features, the invention is not to be limited to that which is described or indicated as contemplated with respect to each variation of the invention. Various changes may be made to the invention described and equivalents (whether recited herein or not included for the sake of some brevity) may be substituted without departing from the true spirit and scope of the invention. In addition, where a range of values is provided, it is understood that every intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention.

Also, it is contemplated that any optional feature of the inventive variations described may be set forth and claimed independently, or in combination with any one or more of the

features described herein. Reference to a singular item, includes the possibility that there are plural of the same items present. More specifically, as used herein and in claims associated hereto, the singular forms "a," "an," "said," and "the" include plural referents unless the specifically stated otherwise. In other words, use of the articles allow for "at least one" of the subject item in the description above as well as claims associated with this disclosure. It is further noted that such claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as "solely," "only" and the like in connection with the recitation of claim elements, or use of a "negative" limitation.

Without the use of such exclusive terminology, the term "comprising" in claims associated with this disclosure shall allow for the inclusion of any additional element--irrespective of whether a given number of elements are enumerated in such claims, or the addition of a feature could be regarded as transforming the nature of an element set forth in such claims. Except as specifically defined herein, all technical and scientific terms used herein are to be given as broad a commonly understood meaning as possible while maintaining claim validity.

The breadth of the present invention is not to be limited to the examples provided and/or the subject specification, but rather only by the scope of claim language associated with this disclosure.

CLAIMS:

1. A system, method, apparatus, or configuration in accordance with any of the aircraft supplemental oxygen embodiments, elements, or combinations thereof described herein.

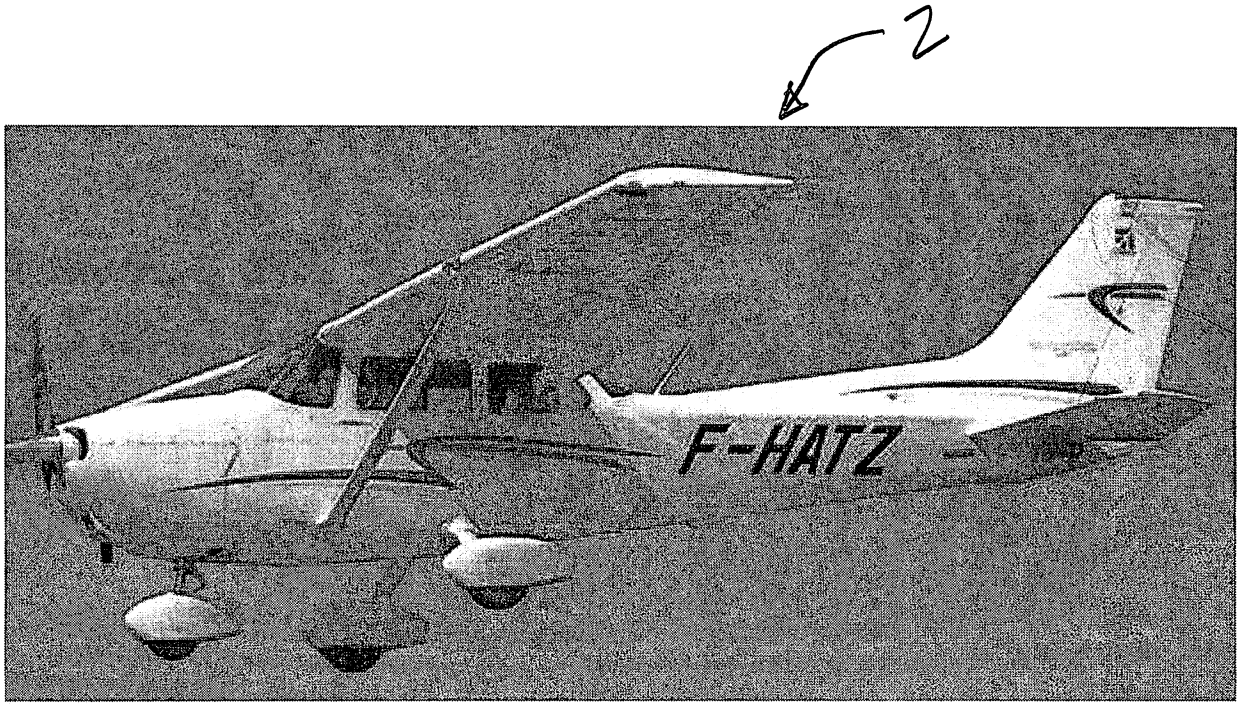


Figure 1A

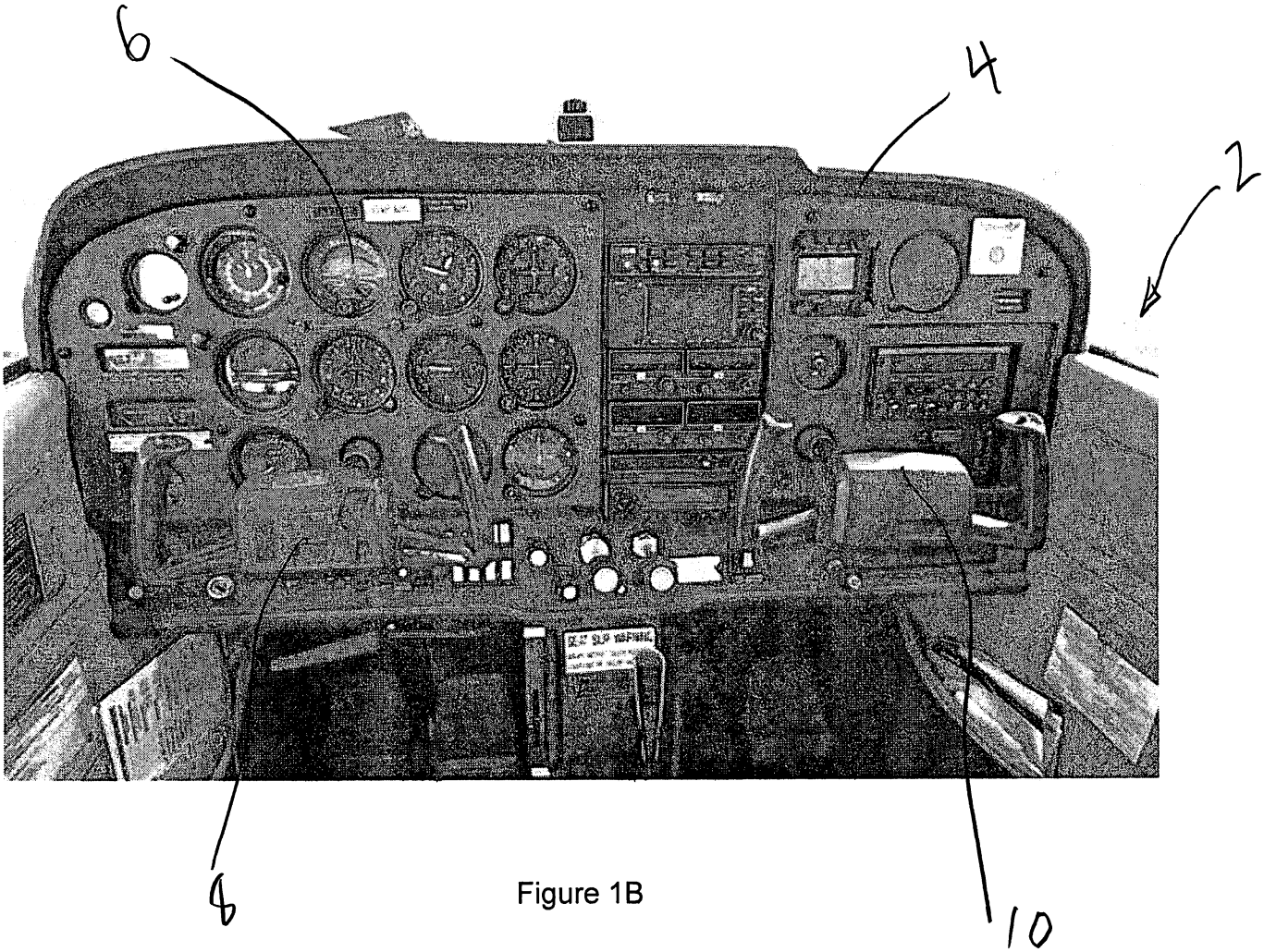


Figure 1B



Figure 2A

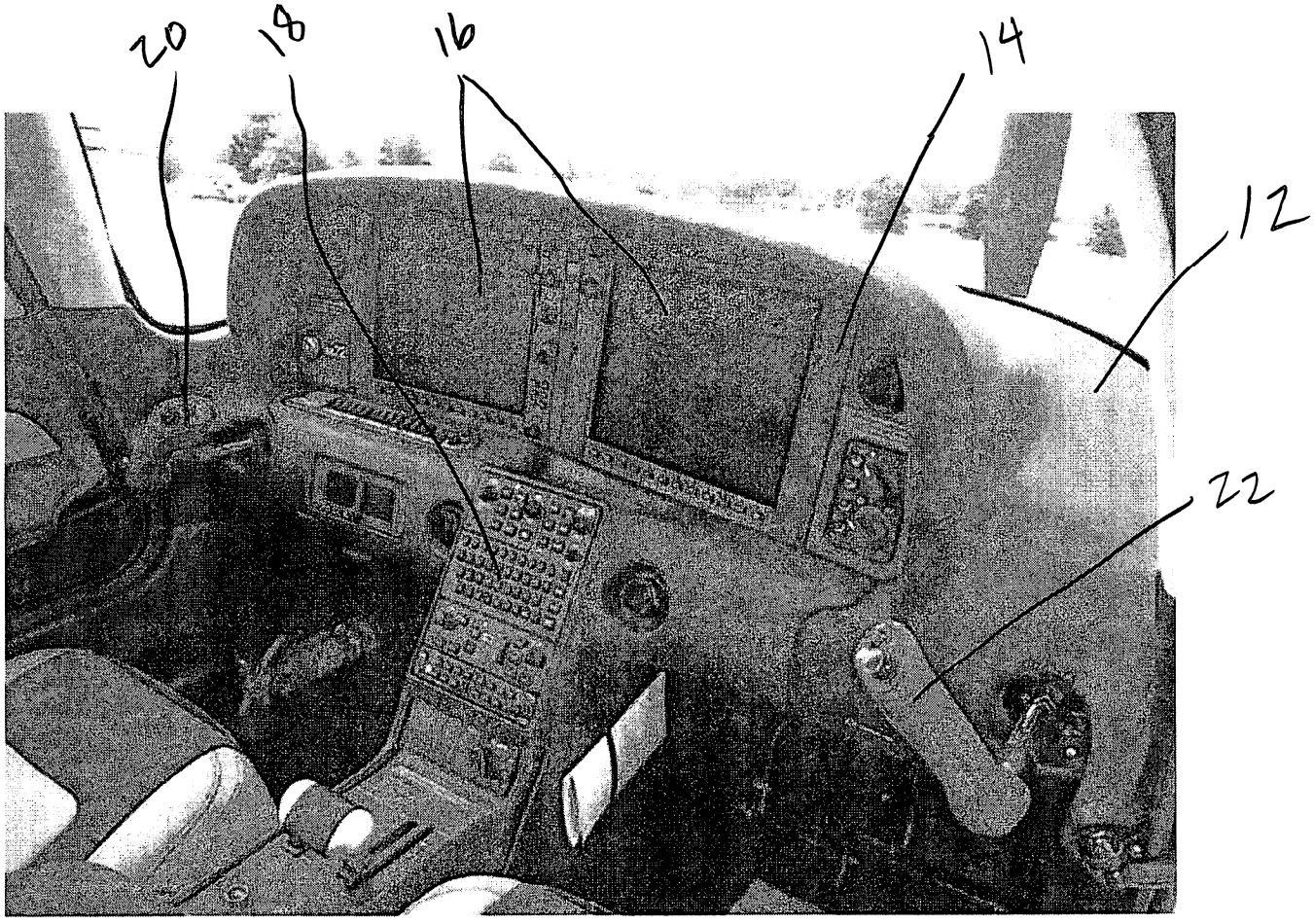


Figure 2B

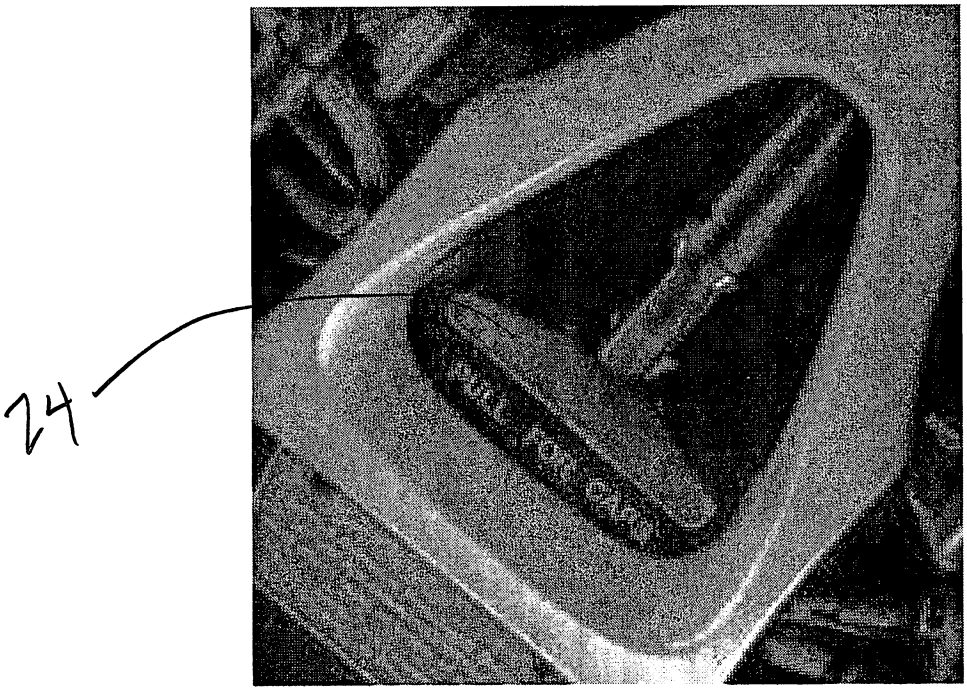


Figure 2C

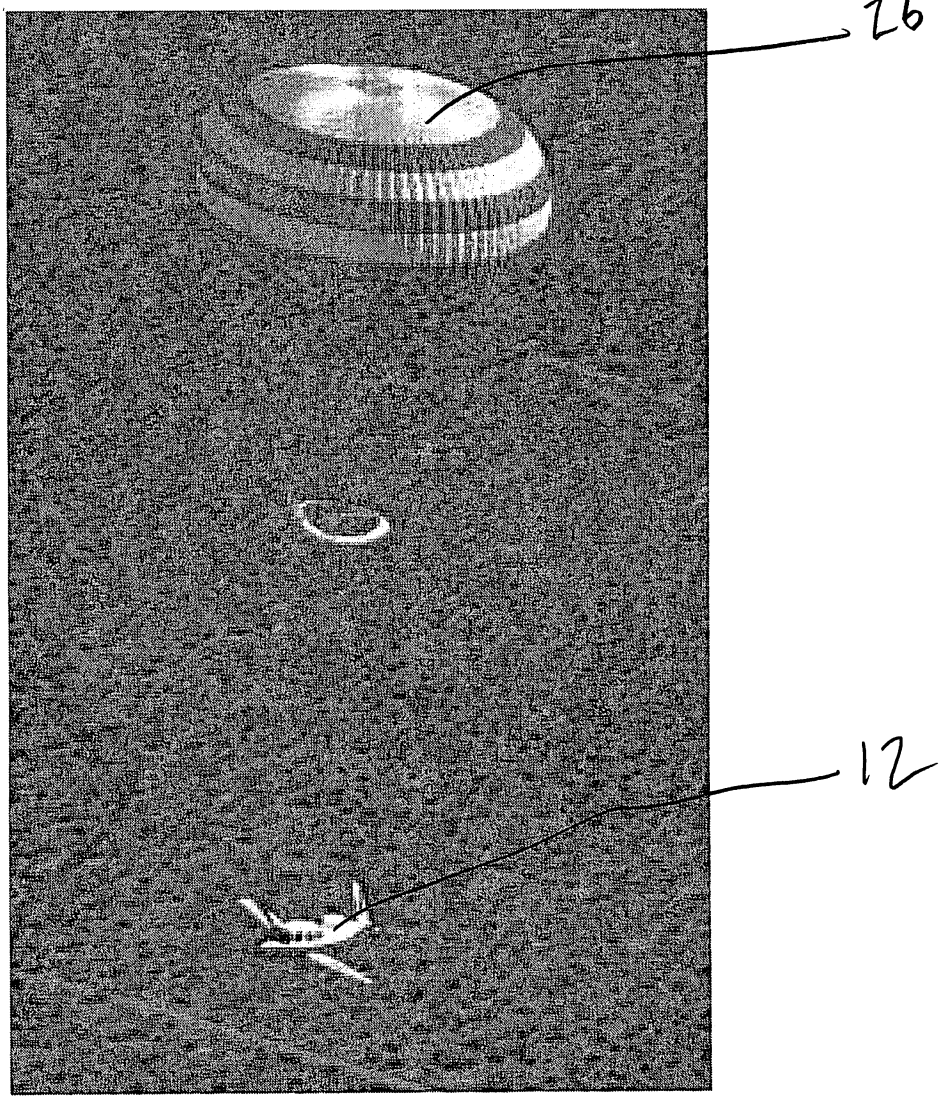


Figure 2D

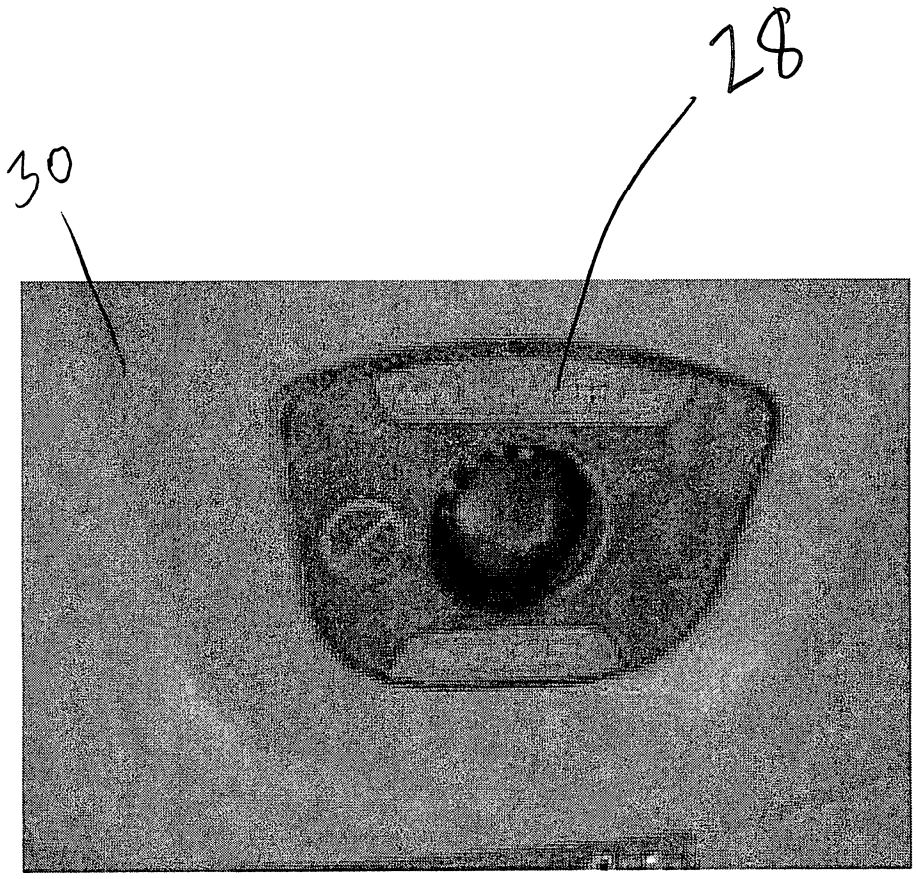


Figure 2E



Figure 3A

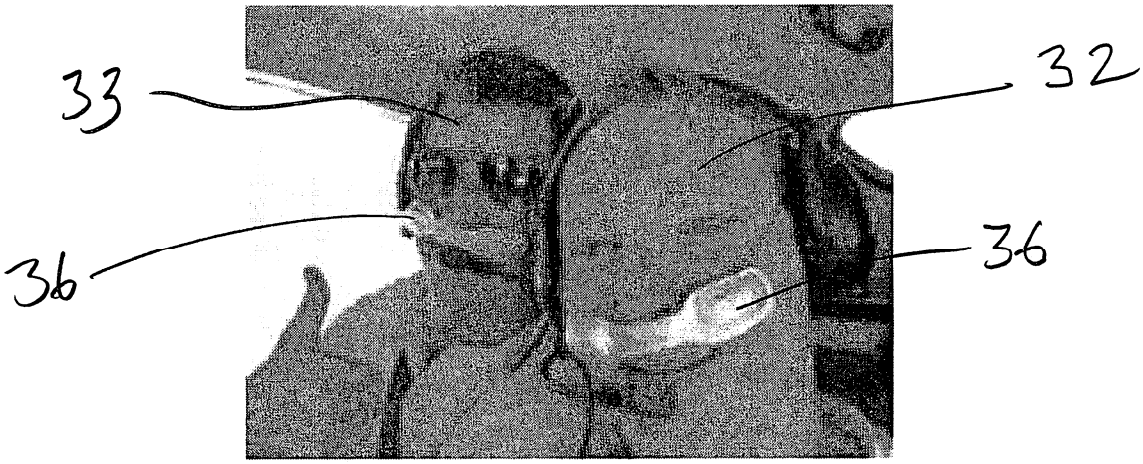


Figure 3B



Figure 3C

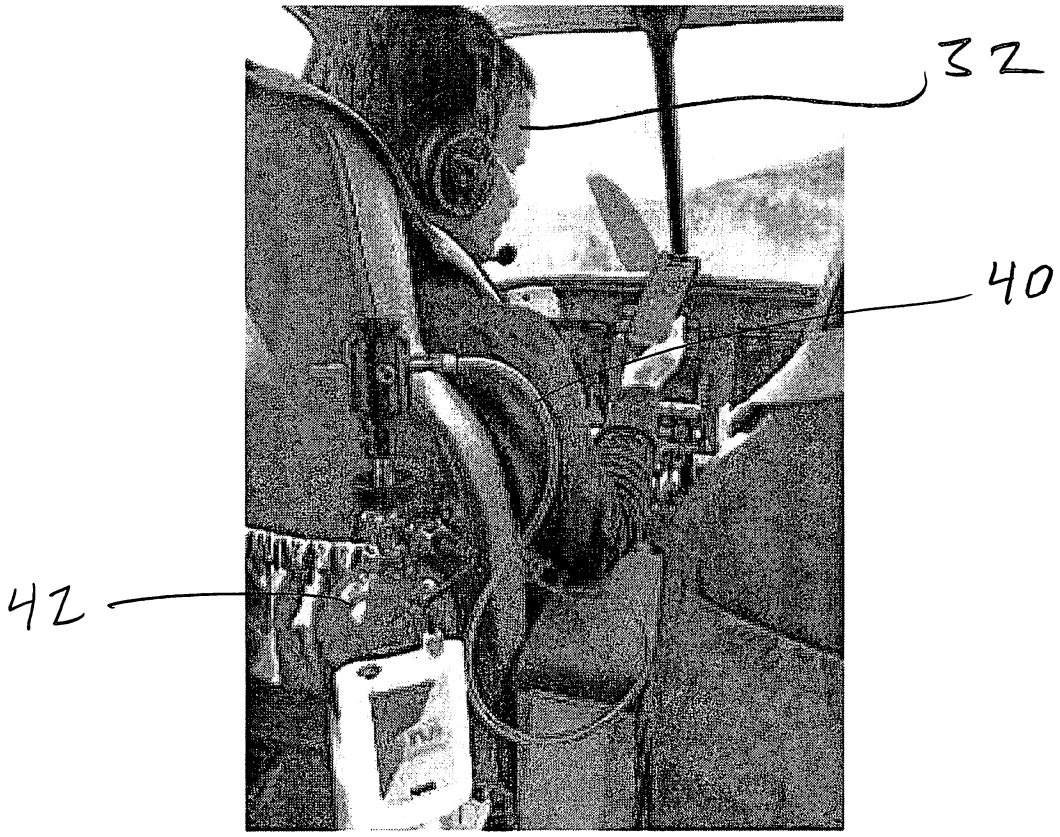


Figure 4A

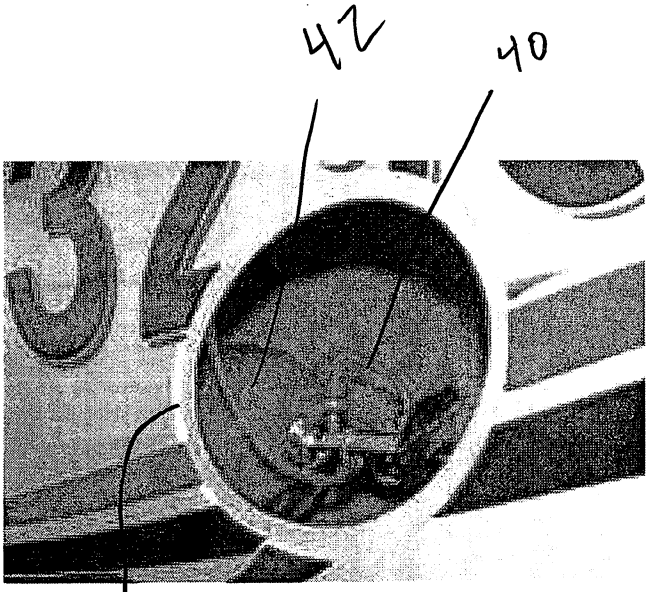


Figure 4B

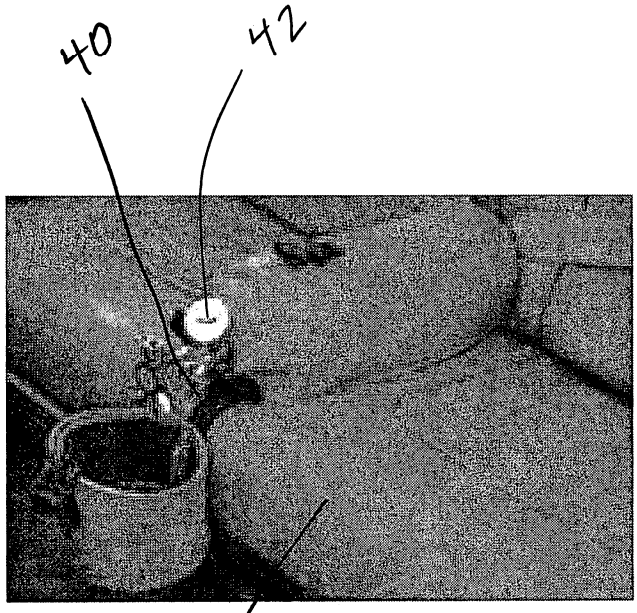


Figure 4C

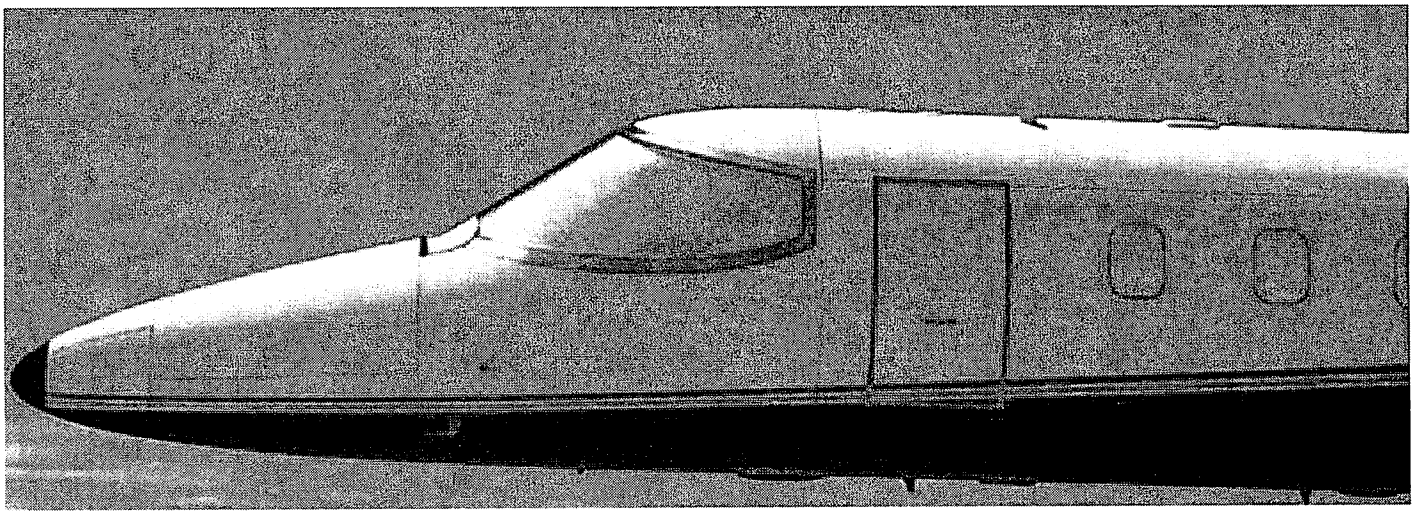


Figure 5A

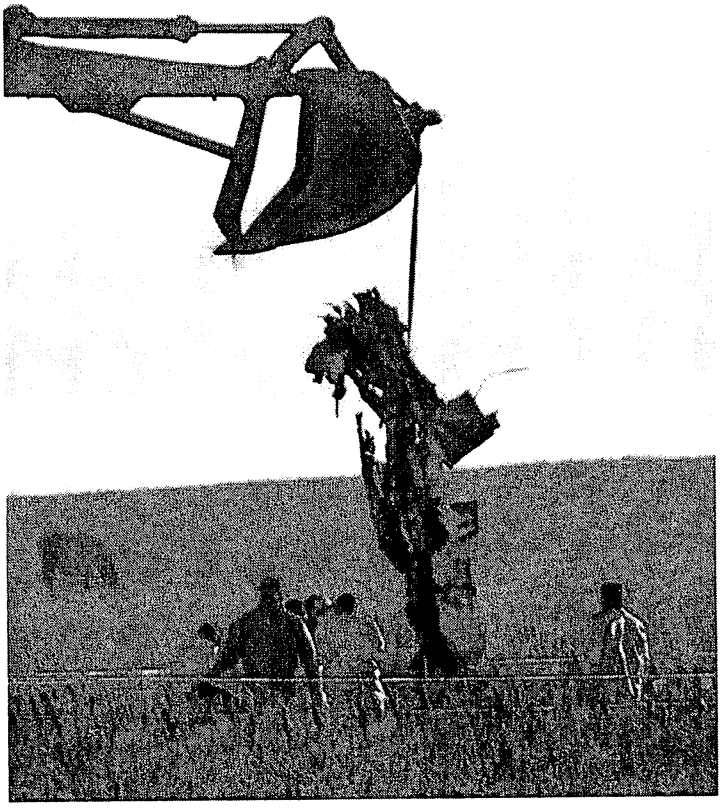


Figure 5B

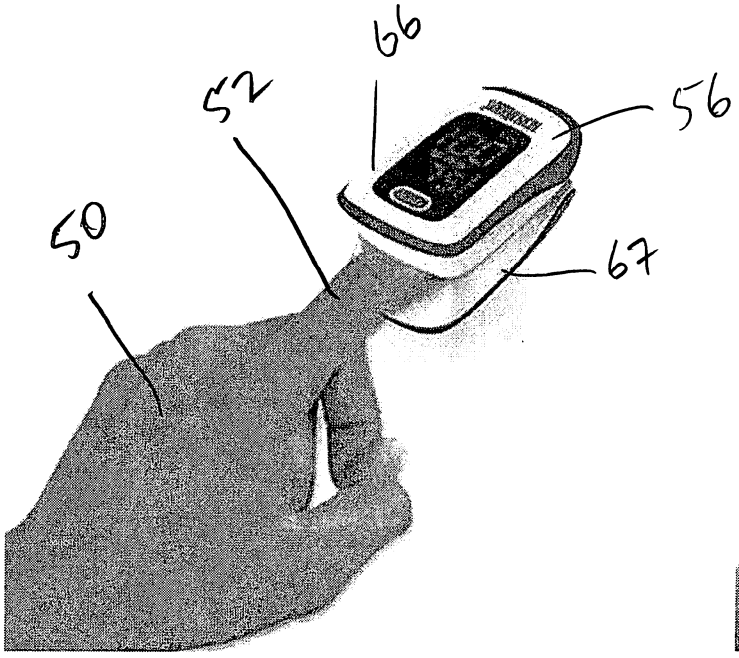


Figure 6A

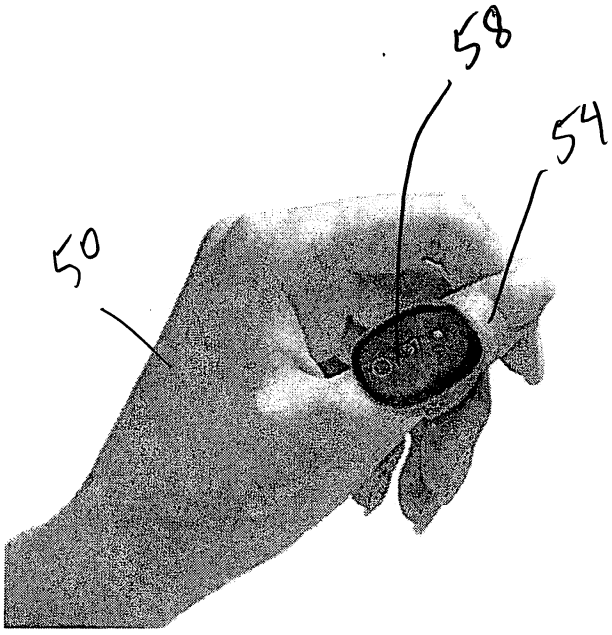


Figure 6B

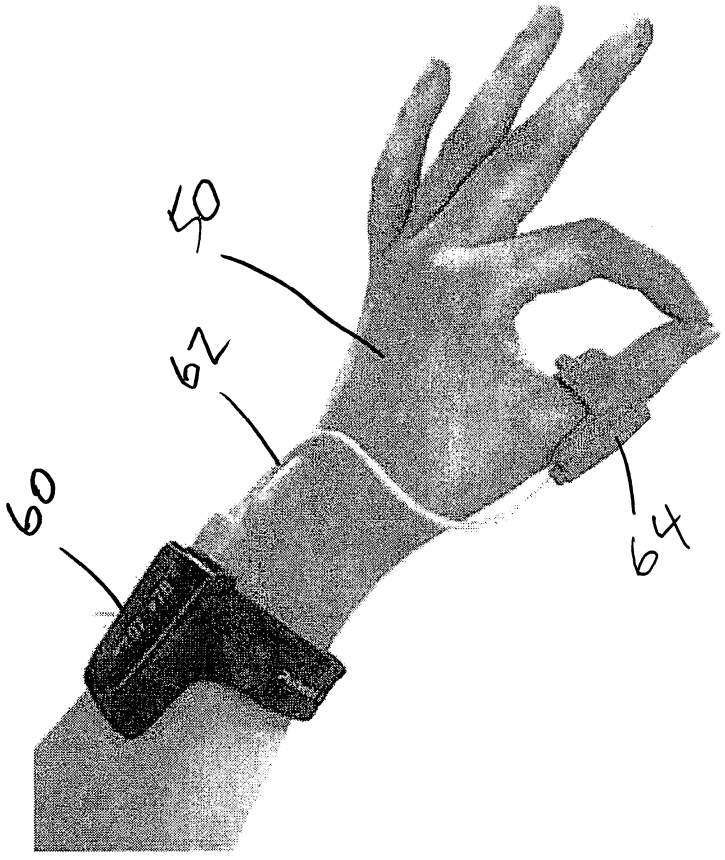


Figure 6C

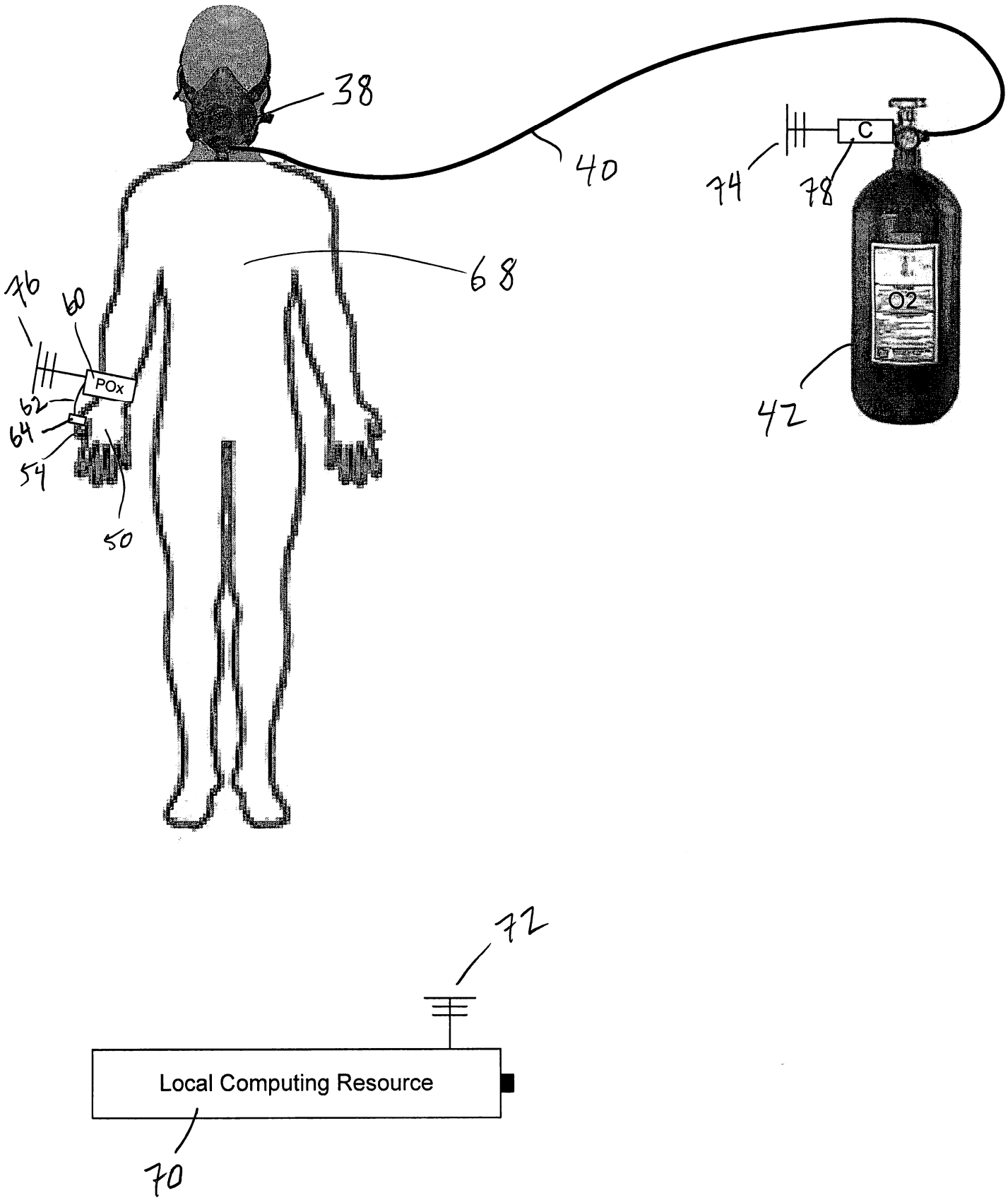

38
40
C
74
78
O2
42
68
76
60
POx
62
64
54
50
72
Local Computing Resource
70

Figure 7A

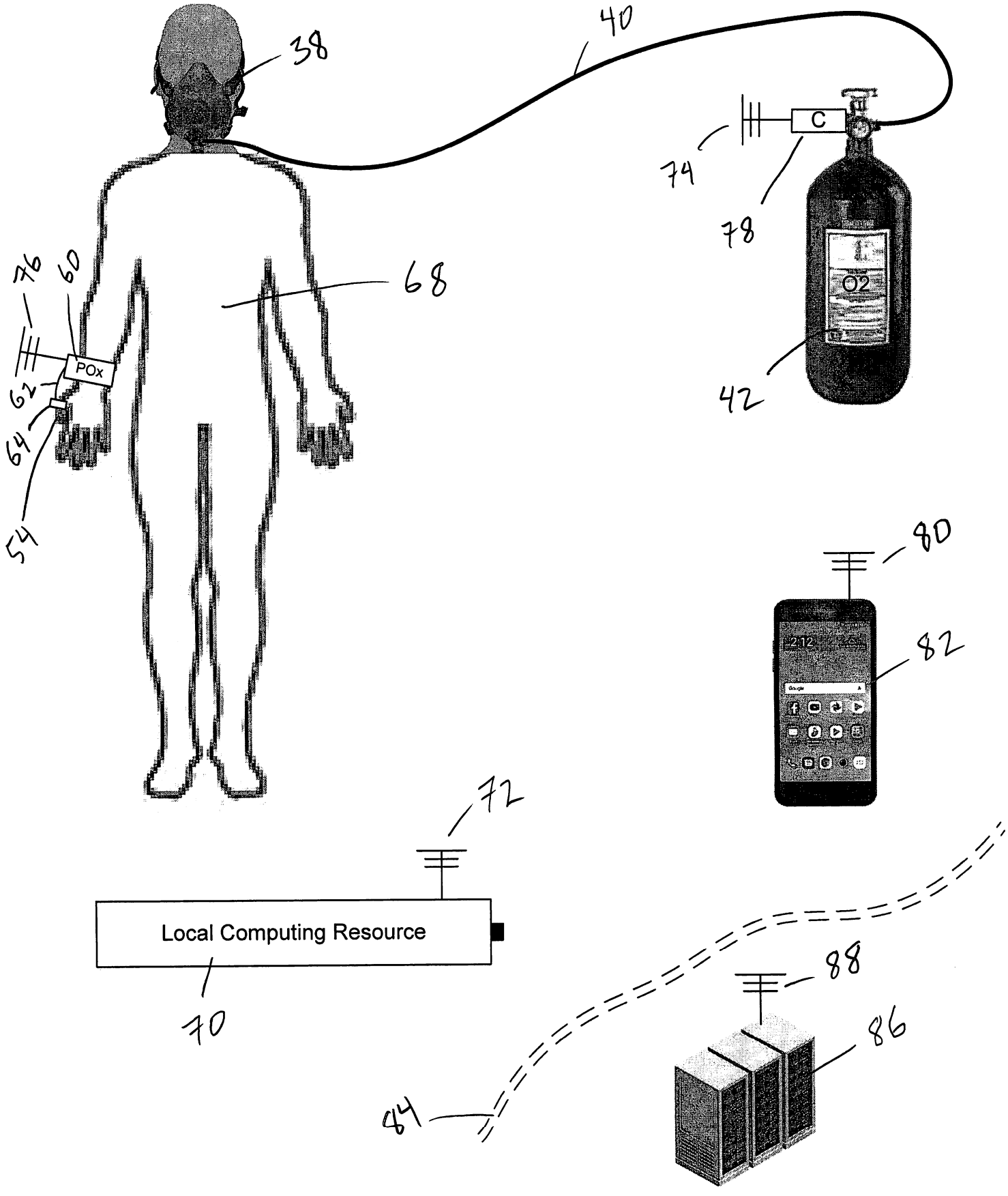

38
40
C
74
78
O2
42
76
60
68
POx
62
64
54
80
82
72
Local Computing Resource
70
88
86
44

Figure 7B

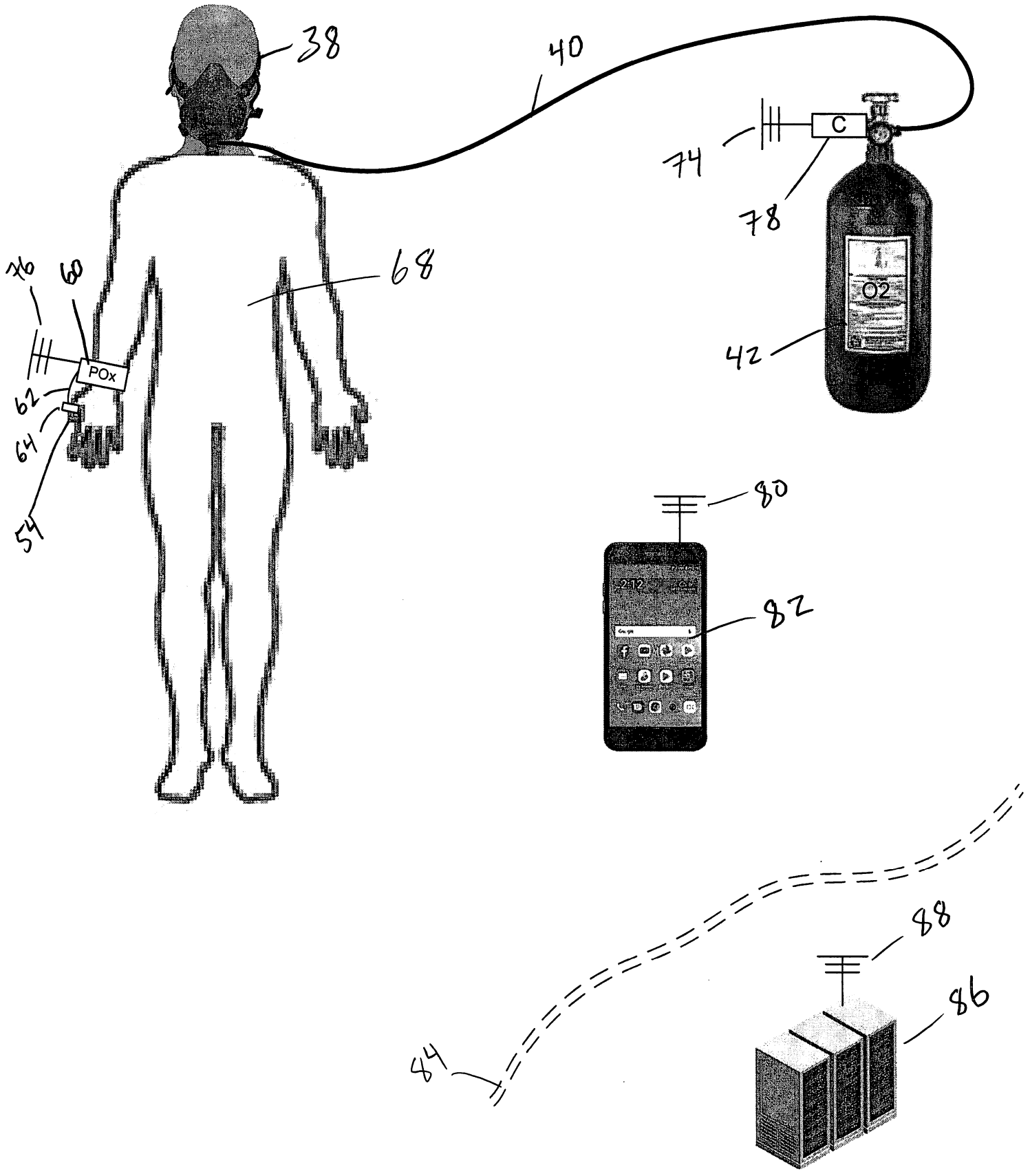


Figure 7C

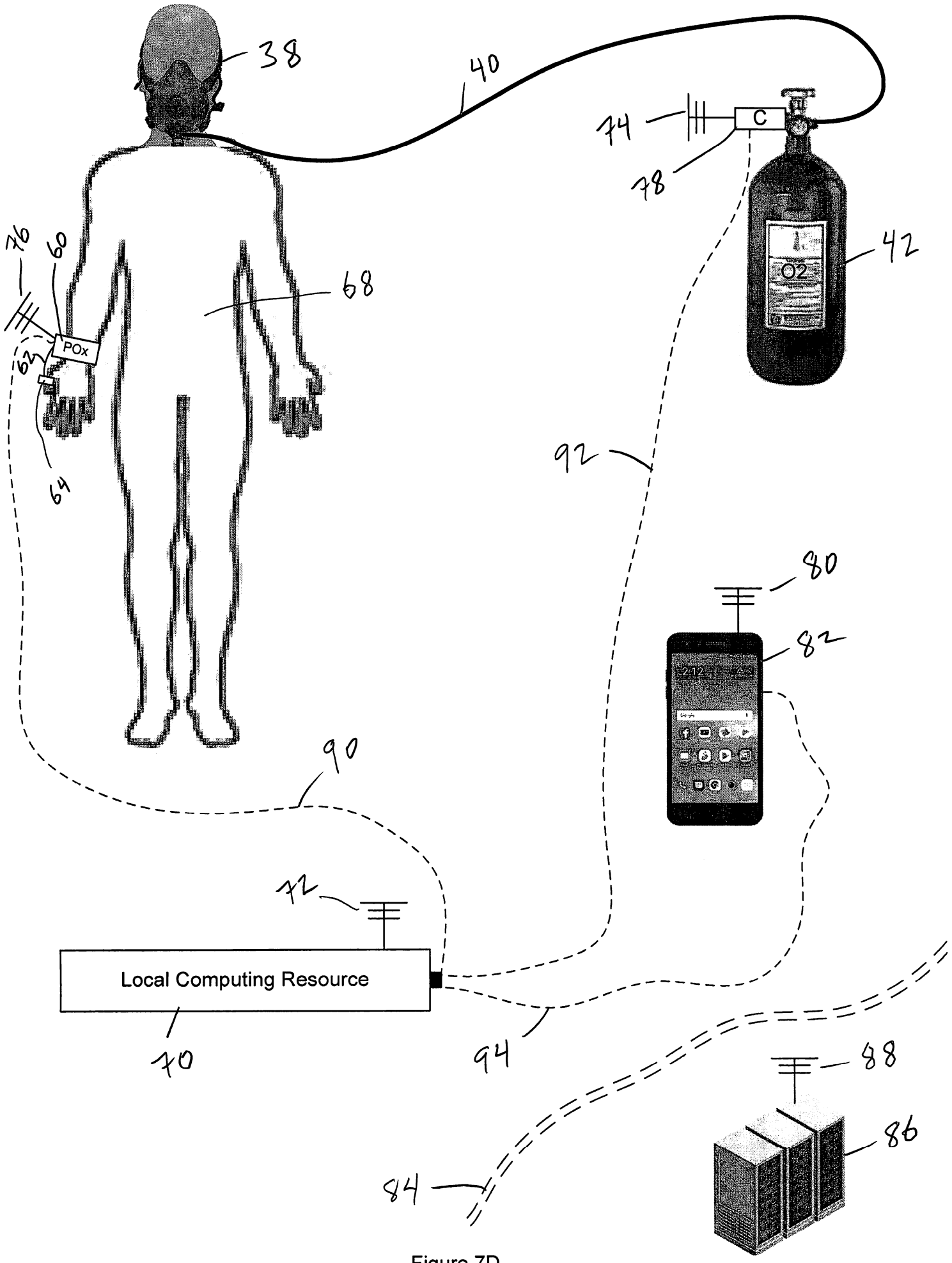

38
40
74
C
78
42
O2
76
60
68
POx
62
64
92
80
82
90
72
Local Computing Resource
70
94
88
86
84

Figure 7D

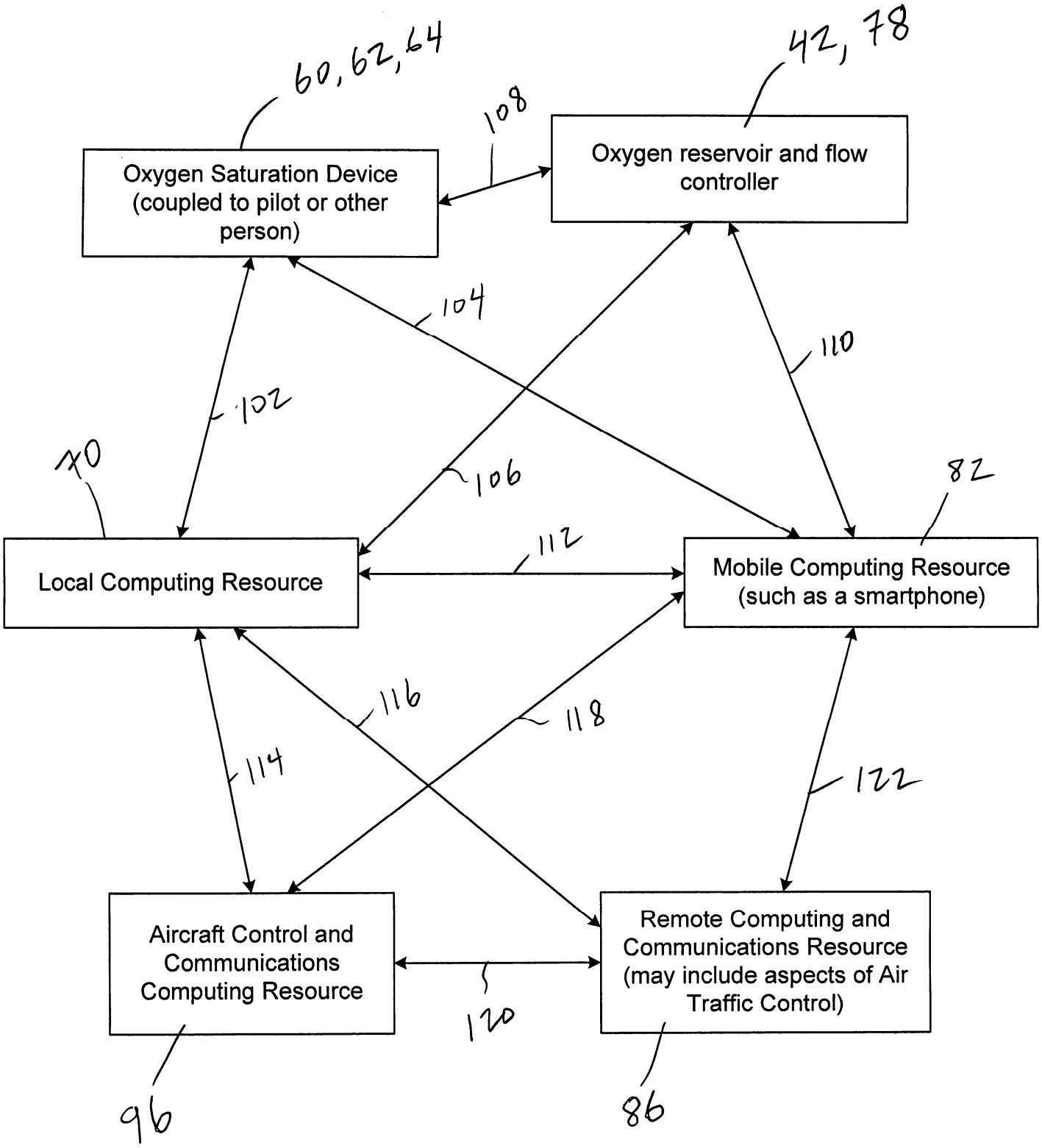

60, 62, 64
42, 78
108
Oxygen Saturation Device (coupled to pilot or other person)
Oxygen reservoir and flow controller
104
110
102
106
40
82
Local Computing Resource
112
Mobile Computing Resource (such as a smartphone)
116
118
114
122
Aircraft Control and Communications Computing Resource
Remote Computing and Communications Resource (may include aspects of Air Traffic Control)
120
96
86

Figure 8

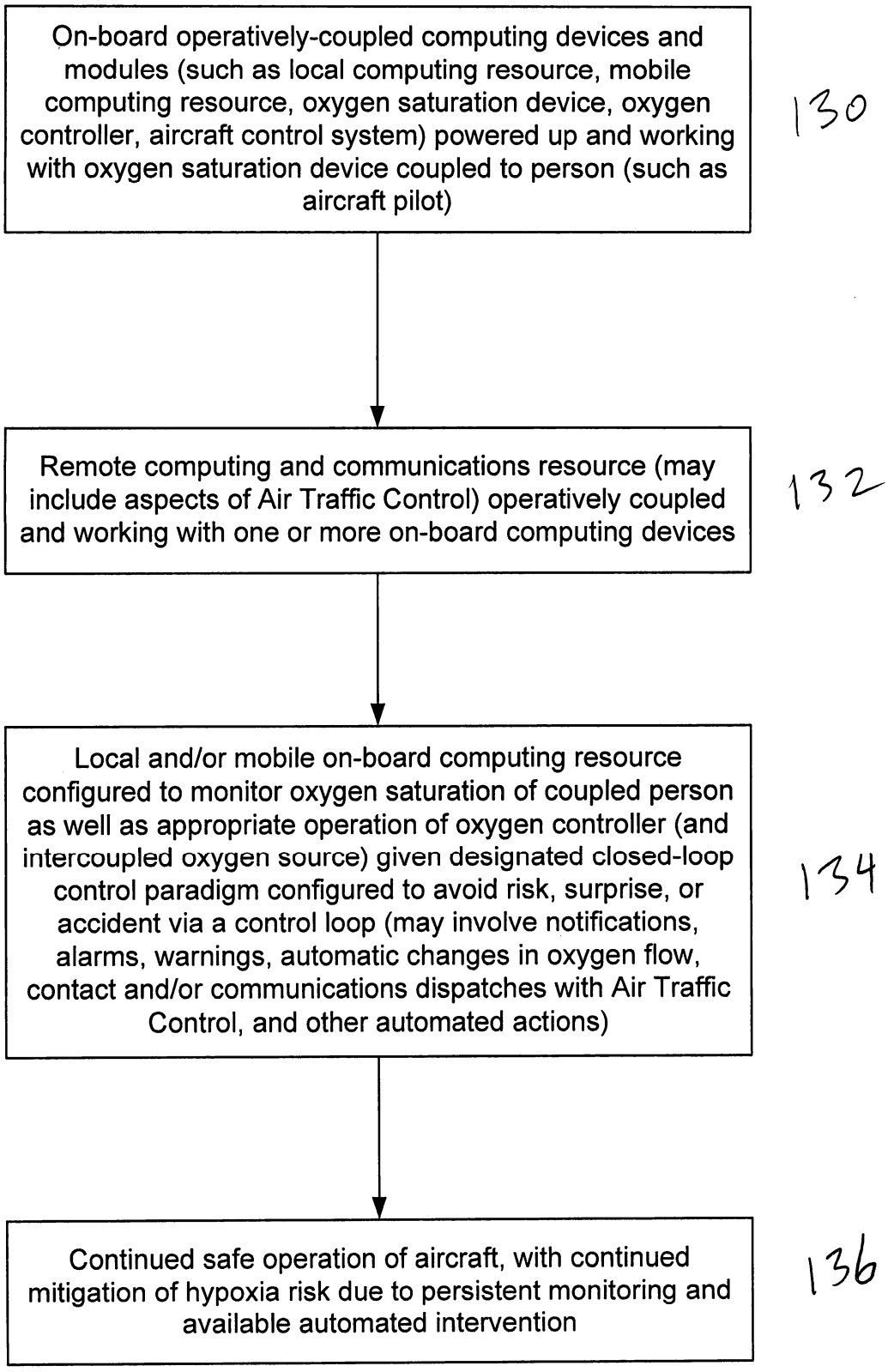

On-board operatively-coupled computing devices and modules (such as local computing resource, mobile computing resource, oxygen saturation device, oxygen controller, aircraft control system) powered up and working with oxygen saturation device coupled to person (such as aircraft pilot)
130
Remote computing and communications resource (may include aspects of Air Traffic Control) operatively coupled and working with one or more on-board computing devices
132
Local and/or mobile on-board computing resource configured to monitor oxygen saturation of coupled person as well as appropriate operation of oxygen controller (and intercoupled oxygen source) given designated closed-loop control paradigm configured to avoid risk, surprise, or accident via a control loop (may involve notifications, alarms, warnings, automatic changes in oxygen flow, contact and/or communications dispatches with Air Traffic Control, and other automated actions)
134
Continued safe operation of aircraft, with continued mitigation of hypoxia risk due to persistent monitoring and available automated intervention
136

Figure 9

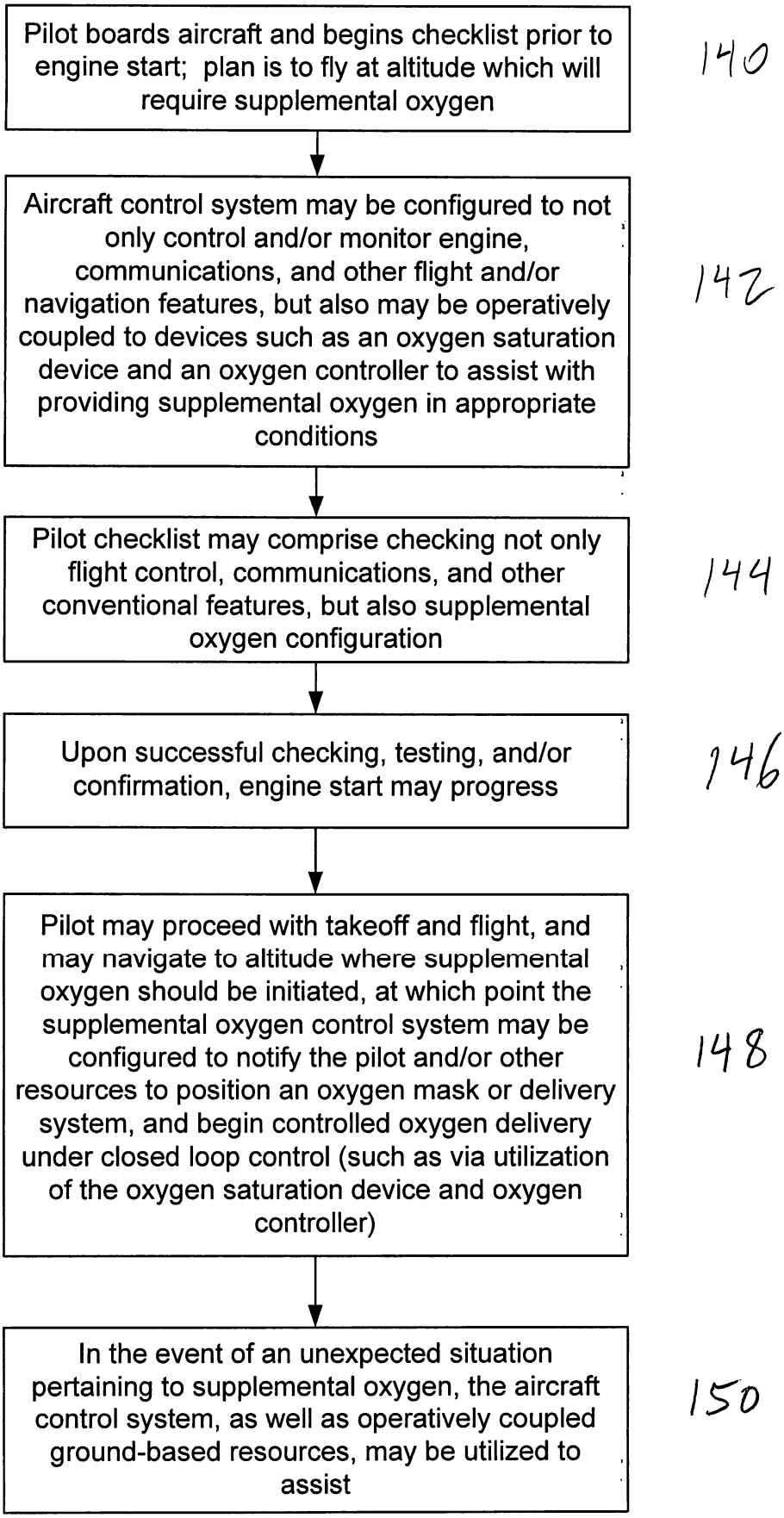

Pilot boards aircraft and begins checklist prior to engine start; plan is to fly at altitude which will require supplemental oxygen
140
Aircraft control system may be configured to not only control and/or monitor engine, communications, and other flight and/or navigation features, but also may be operatively coupled to devices such as an oxygen saturation device and an oxygen controller to assist with providing supplemental oxygen in appropriate conditions
142
Pilot checklist may comprise checking not only flight control, communications, and other conventional features, but also supplemental oxygen configuration
144
Upon successful checking, testing, and/or confirmation, engine start may progress
146
Pilot may proceed with takeoff and flight, and may navigate to altitude where supplemental oxygen should be initiated, at which point the supplemental oxygen control system may be configured to notify the pilot and/or other resources to position an oxygen mask or delivery system, and begin controlled oxygen delivery under closed loop control (such as via utilization of the oxygen saturation device and oxygen controller)
148
In the event of an unexpected situation pertaining to supplemental oxygen, the aircraft control system, as well as operatively coupled ground-based resources, may be utilized to assist
150

Figure 10